5. Sieci komputerowe. Część 1 - Homework

192.168.10.37/24

Adres IP: 192.168.10.37 Prefix: /24

IP binarnie: 11000000.10101000.00001010.00100101

Maska binarnie: 11111111.11111111.11111111.00000000

Maska dziesiętnie: 255.255.255.0

NETWORK | HOST: 11000000.10101000.00001010.|00100101

Network Address: 192.168.10.0

Pierwszy host: 192.168.10.1

Ostatni host: 192.168.10.254

Broadcast: 192.168.10.255

Liczba adresów: 256

Liczba użytecznych hostów: 254

Adres IP: 192.168.10.0 Prefix: /25

IP binarnie: 11000000.10101000.00001010.00000000

Maska binarnie: 11111111.11111111.11111111.10000000

Maska dziesiętnie: 255.255.255.128

NETWORK | HOST: 11000000.10101000.00001010.0 | 0000000

Network Address: 192.168.10.0

Pierwszy host: 192.168.10.1

Ostatni host: 192.168.10.126

Broadcast: 192.168.10.127

Liczba adresów: 128 (2^7)

Liczba użytecznych hostów: 126 (2^7)-2

Adres IP: 192.168.10.146 Prefix: /26

IP binarnie: 11000000.10101000.00001010.10010010

Maska binarnie: 11111111.11111111.11111111.11000000

Maska dziesiętnie: 255.255.255.192

NETWORK | HOST: 11000000.10101000.00001010.10 | 010010

Network Address: 192.168.10.128

Pierwszy host: 192.168.10.129

Ostatni host: 192.168.10.190

Broadcast: 192.168.10.191

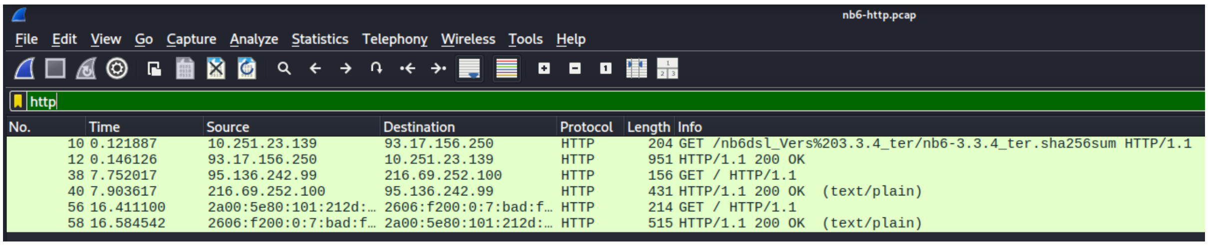
Liczba adresów: 64

Liczba użytecznych hostów: 62

1. **Enkapsulacja i model OSI**

Znajdź pakiet zawierający żądanie HTTP GET i rozwiń jego strukturę w panelu szczegółów Wiresharka. Jakiego filtru użyjesz?

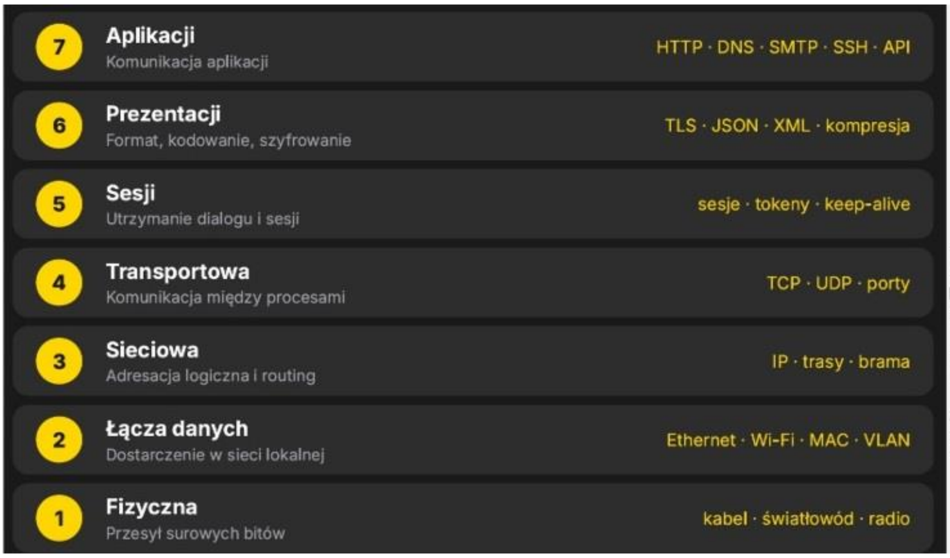
Filtr “HTTP”, "GET", ewentualnie “tcp.port == 80” Wyniki: No 10, 38, 214

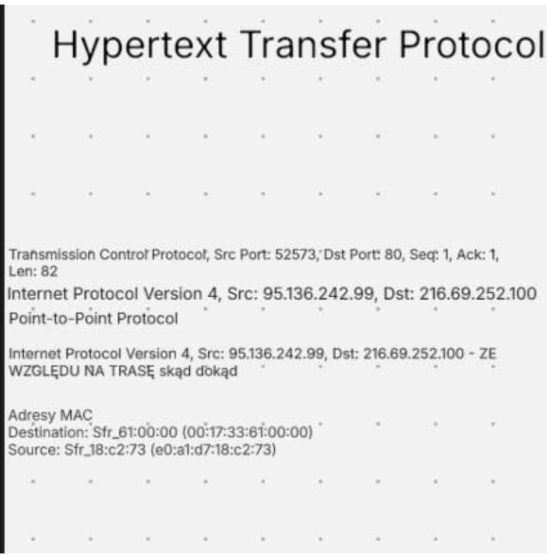


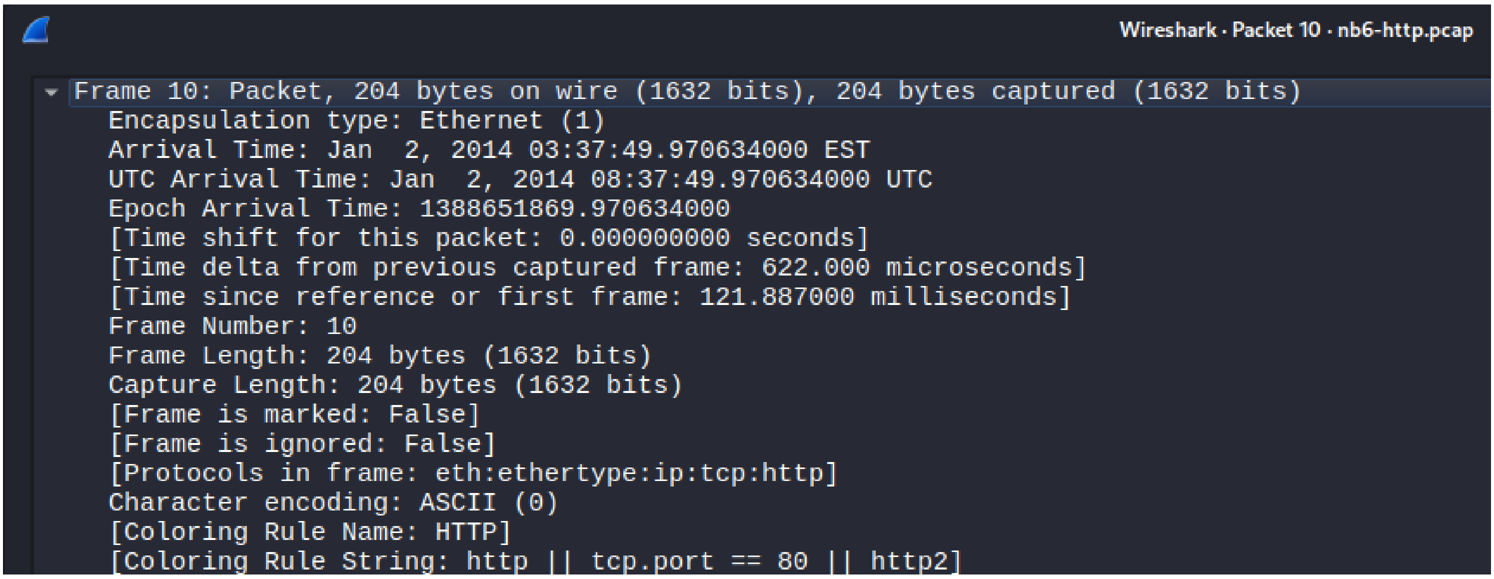
Jakie protokoły reprezentują warstwy 2, 3, 4 i 7 modelu OSI?

Frame 10: Packet, 204 bytes on wire (1632 bits), 204 bytes captured (1632 bits)

- Warstwa 1





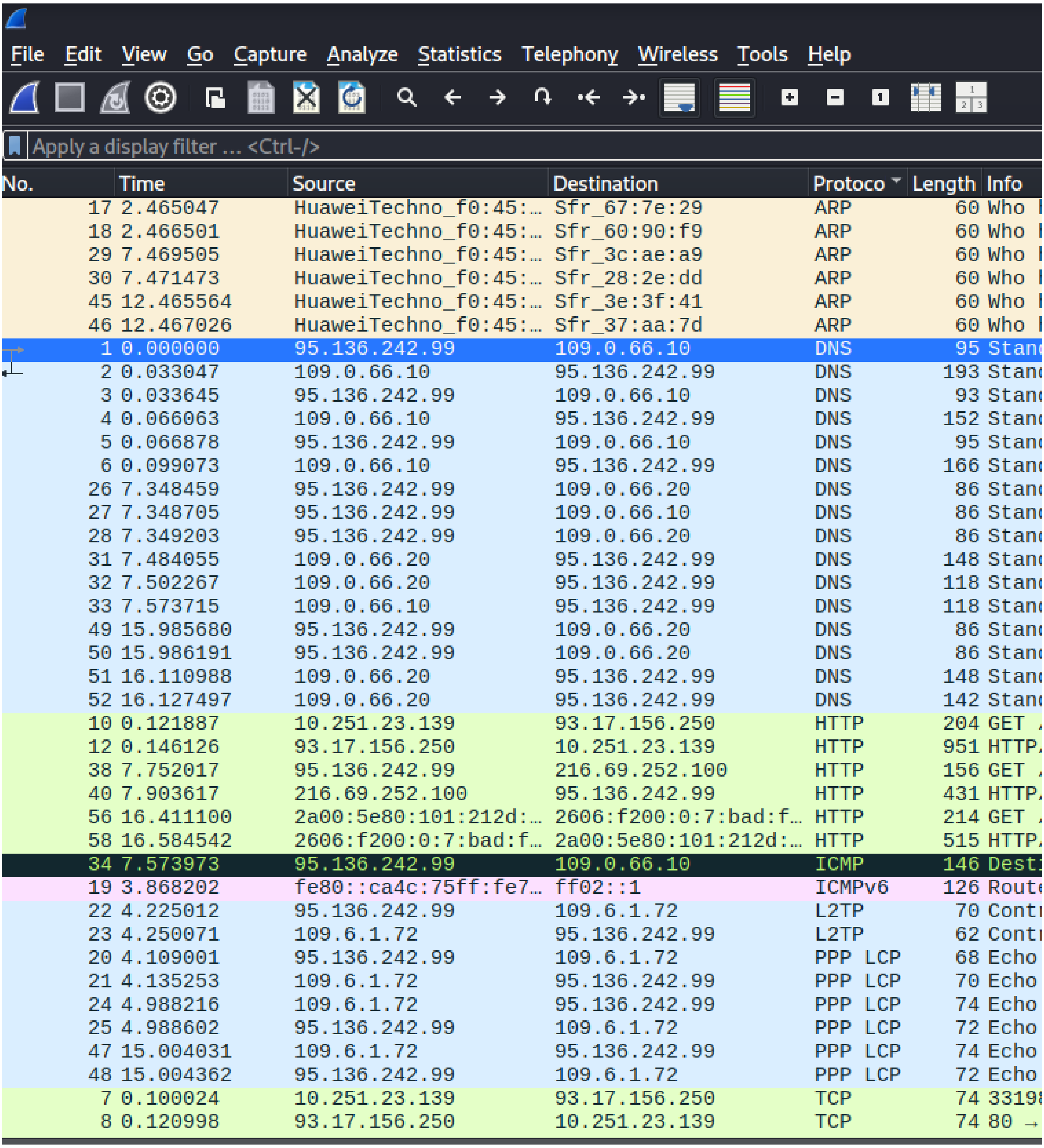


TCP - 4 transportowej ICMP - 3 Sieciowa HTTP - 7 Aplikacji DNS - 7 Aplikacji PPP LCP - 2 Warstwa Danych L2TP - 2 Warstwa Danych

Jak nazywa się jednostka danych na każdej z tych warstw?

2. ramki 3. pakiety 4. segment 7. data

Jakie protokoły występują w analizowanym pliku?



No.	Time	Source	Destination	Protoco	Length	Info
17	2.465047	HuaweiTechno_f0:45:…	Sfr_67:7e:29	ARP	60	Who I
18	2.466501	HuaweiTechno_f0:45:…	Sfr_60:90:f9	ARP	60	Who I
29	7.469505	HuaweiTechno_f0:45:…	Sfr_3c:ae:a9	ARP	60	Who I
30	7.471473	HuaweiTechno_f0:45:…	Sfr_28:2e:dd	ARP	60	Who I
45	12.465564	HuaweiTechno_f0:45:…	Sfr_3e:3f:41	ARP	60	Who I
46	12.467026	HuaweiTechno_f0:45:…	Sfr_37:aa:7d	ARP	60	Who I
1	0.000000	95.136.242.99	109.0.66.10	DNS	95	Stan
2	0.033047	109.0.66.10	95.136.242.99	DNS	193	Stan
3	0.033645	95.136.242.99	109.0.66.10	DNS	93	Stan
4	0.066063	109.0.66.10	95.136.242.99	DNS	152	Stan
5	0.066878	95.136.242.99	109.0.66.10	DNS	95	Stan
6	0.099073	109.0.66.10	95.136.242.99	DNS	166	Stan
26	7.348459	95.136.242.99	109.0.66.20	DNS	86	Stan
27	7.348705	95.136.242.99	109.0.66.10	DNS	86	Stan
28	7.349203	95.136.242.99	109.0.66.20	DNS	86	Stan
31	7.484055	109.0.66.20	95.136.242.99	DNS	148	Stan
32	7.502267	109.0.66.20	95.136.242.99	DNS	118	Stan
33	7.573715	109.0.66.10	95.136.242.99	DNS	118	Stan
49	15.985680	95.136.242.99	109.0.66.20	DNS	86	Stan
50	15.986191	95.136.242.99	109.0.66.20	DNS	86	Stan
51	16.110988	109.0.66.20	95.136.242.99	DNS	148	Stan
52	16.127497	109.0.66.20	95.136.242.99	DNS	142	Stan
10	0.121887	10.251.23.139	93.17.156.250	HTTP	204	GET /
12	0.146126	93.17.156.250	10.251.23.139	HTTP	951	HTTP/
38	7.752017	95.136.242.99	216.69.252.100	HTTP	156	GET /
40	7.903617	216.69.252.100	95.136.242.99	HTTP	431	HTTP/
56	16.411100	2a00:5e80:101:212d:…	2606:f200:0:7:bad:f…	HTTP	214	GET /
58	16.584542	2606:f200:0:7:bad:f…	2a00:5e80:101:212d:…	HTTP	515	HTTP/
34	7.573973	95.136.242.99	109.0.66.10	ICMP	146	Dest
19	3.868202	fe80::ca4c:75ff:fe7…	ff02::1	ICMPv6	126	Route
22	4.225012	95.136.242.99	109.6.1.72	L2TP	70	Contr
23	4.250071	109.6.1.72	95.136.242.99	L2TP	62	Contr
20	4.109001	95.136.242.99	109.6.1.72	PPP LCP	68	Echo
21	4.135253	109.6.1.72	95.136.242.99	PPP LCP	70	Echo
24	4.988216	109.6.1.72	95.136.242.99	PPP LCP	74	Echo
25	4.988602	95.136.242.99	109.6.1.72	PPP LCP	72	Echo
47	15.004031	109.6.1.72	95.136.242.99	PPP LCP	74	Echo
48	15.004362	95.136.242.99	109.6.1.72	PPP LCP	72	Echo
7	0.100024	10.251.23.139	93.17.156.250	TCP	74	3319
8	0.120998	93.17.156.250	10.251.23.139	TCP	74	80 →

ARP, DNS, HTTP, ICMP, ICMPv6, L2TP, PPP LCP, TCP

Homework z 13.08.2026

Przejście przez plik pobrany nb6.http ciąg dalszy…

2. Zestawienie połączenia TCP

Znajdź trzy pakiety tworzące TCP three-way handshake.

Jakie flagi występują kolejno? Który host rozpoczyna połączenie? Z jakiego portu klienta i do jakiego portu serwera wykonywane jest połączenie?

Filtr: tcp.flags.syn == 1

```
7 0.100024   10.251.23.139   93.17.156.250   TCP   74 33198 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=550051 TSecr=0 WS=2
8 0.120998   93.17.156.250   10.251.23.139   TCP   74 80 → 33198 [SYN, ACK] Seq=0 Ack=1 Win=14480 Len=0 MSS=1440 SACK_PERM TSval=2307889920 TSecr=550051 WS=512
9 0.121265   10.251.23.139   93.17.156.250   TCP   66 33198 → 80 [ACK] Seq=1 Ack=1 Win=5840 Len=0 TSval=550073 TSecr=2307889920
```

W pakiecie **7**:

33198 → 80 **[SYN]** Seq=0

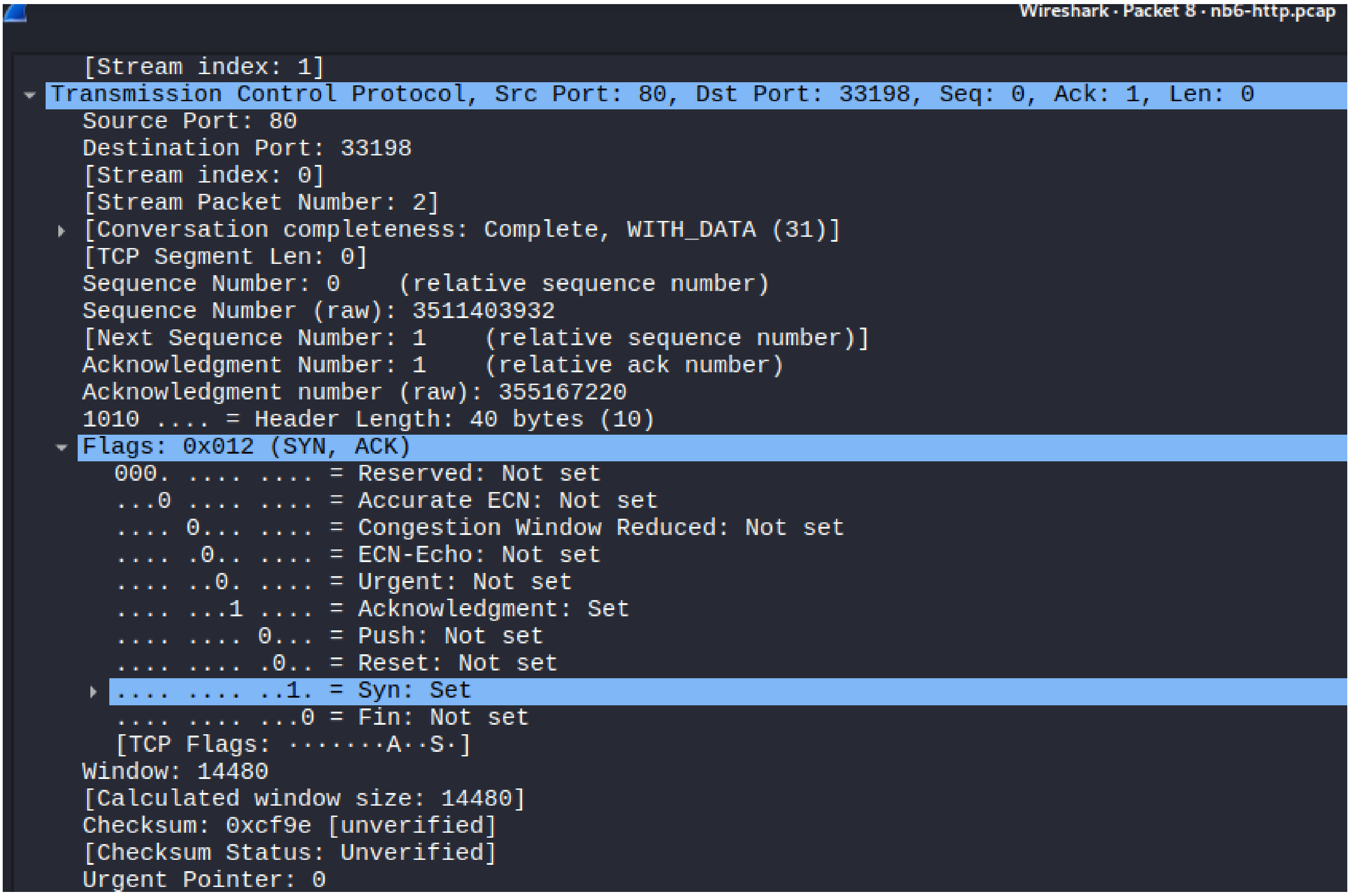
```
Wireshark · Packet 7 · nb6-http.pcap
  Header Checksum: 0x0e27 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 10.251.23.139
  Destination Address: 93.17.156.250
  [Stream index: 1]
Transmission Control Protocol, Src Port: 33198, Dst Port: 80, Seq: 0, Len: 0
  Source Port: 33198
  Destination Port: 80
  [Stream index: 0]
  [Stream Packet Number: 1]
▸ [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 0    (relative sequence number)
  Sequence Number (raw): 355167219
  [Next Sequence Number: 1    (relative sequence number)]
  Acknowledgment Number: 0
  Acknowledgment number (raw): 0
  1010 .... = Header Length: 40 bytes (10)
▾ Flags: 0x002 (SYN)
    000. .... .... = Reserved: Not set
    ...0 .... .... = Accurate ECN: Not set
    .... 0... .... = Congestion Window Reduced: Not set
    .... .0.. .... = ECN-Echo: Not set
    .... ..0. .... = Urgent: Not set
    .... ...0 .... = Acknowledgment: Not set
    .... .... 0... = Push: Not set
    .... .... .0.. = Reset: Not set
  ▸ .... .... ..1. = Syn: Set
    .... .... ...0 = Fin: Not set
    [TCP Flags: ··········S·]
  Window: 5840
  [Calculated window size: 5840]
  Checksum: 0xacdc [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
▸ Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
▾ [Timestamps]
    [Time since first frame in this TCP stream: 0.000000000 seconds]
    [Time since previous frame in this TCP stream: 0.000000000 seconds]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
```

- port źródłowy: **33198**

- port docelowy: **80**
- flaga **SYN = 1**
- Seq=0

W pakiecie **8**:

80 → 33198 **[SYN, ACK]** Seq=0 Ack=1



```
Wireshark · Packet 8 · nb6-http.pcap
    [Stream index: 1]
  Transmission Control Protocol, Src Port: 80, Dst Port: 33198, Seq: 0, Ack: 1, Len: 0
    Source Port: 80
    Destination Port: 33198
    [Stream index: 0]
    [Stream Packet Number: 2]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 3511403932
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 355167220
    1010 .... = Header Length: 40 bytes (10)
    Flags: 0x012 (SYN, ACK)
      000. .... .... = Reserved: Not set
      ...0 .... .... = Accurate ECN: Not set
      .... 0... .... = Congestion Window Reduced: Not set
      .... .0.. .... = ECN-Echo: Not set
      .... ..0. .... = Urgent: Not set
      .... ...1 .... = Acknowledgment: Set
      .... .... 0... = Push: Not set
      .... .... .0.. = Reset: Not set
      .... .... ..1. = Syn: Set
      .... .... ...0 = Fin: Not set
      [TCP Flags: ·······A··S·]
    Window: 14480
    [Calculated window size: 14480]
    Checksum: 0xcf9e [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
```

- serwer odpowiada z portu **80**
- **SYN = 1**
- **ACK = 1**
- Ack=1 oznacza potwierdzenie otrzymania pierwszego numeru sekwencyjnego klienta.

W pakiecie **9**:

33198 → 80 **[ACK]** Seq=1 Ack=1

```
Wireshark · Packet 9 · nb6-http.pcap

Transmission Control Protocol, Src Port: 33198, Dst Port: 80, Seq: 1, Ack: 1, Len: 0
    Source Port: 33198
    Destination Port: 80
    [Stream index: 0]
    [Stream Packet Number: 3]
  > [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 355167220
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 3511403933
    1000 .... = Header Length: 32 bytes (8)
  v Flags: 0x010 (ACK)
      000. .... .... = Reserved: Not set
      ...0 .... .... = Accurate ECN: Not set
      .... 0... .... = Congestion Window Reduced: Not set
      .... .0.. .... = ECN-Echo: Not set
      .... ..0. .... = Urgent: Not set
      .... ...1 .... = Acknowledgment: Set
      .... .... 0... = Push: Not set
      .... .... .0.. = Reset: Not set
      .... .... ..0. = Syn: Not set
      .... .... ...0 = Fin: Not set
      [TCP Flags: ·······A····]
    Window: 2920
    [Calculated window size: 5840]
    [Window size scaling factor: 2]
    Checksum: 0x2b6b [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  > Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  v [Timestamps]
      [Time since first frame in this TCP stream: 21.241000 milliseconds]
      [Time since previous frame in this TCP stream: 267.000 microseconds]
  v [SEQ/ACK analysis]
      [This is an ACK to the segment in frame: 8]
      [The RTT to ACK the segment was: 267.000 microseconds]
      [iRTT: 21.241000 milliseconds]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
```

- klient wysyła **ACK**
- Ack=1 potwierdza SYN serwera.

Pozostałe three-way handshake:

No.	Time	Source	Destination	Protocol	Length	Info
10	0.121887	10.251.23.139	93.17.156.250	HTTP	204	GET /nb6dsl_Vers%203.3.4_ter/nb6-3.3.4_ter.sha256sum HTTP/1.1
12	0.146126	93.17.156.250	10.251.23.139	HTTP	951	HTTP/1.1 200 OK
38	7.752017	95.136.242.99	216.69.252.100	HTTP	156	GET / HTTP/1.1
40	7.903617	216.69.252.100	95.136.242.99	HTTP	431	HTTP/1.1 200 OK (text/plain)
56	16.411100	2a00:5e80:101:212d:…	2606:f200:0:7:bad:f…	HTTP	214	GET / HTTP/1.1
58	16.584542	2606:f200:0:7:bad:f…	2a00:5e80:101:212d:…	HTTP	515	HTTP/1.1 200 OK (text/plain)
7	0.100024	10.251.23.139	93.17.156.250	TCP	74	33198 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=550051 TSecr=0 WS=2
8	0.120998	93.17.156.250	10.251.23.139	TCP	74	80 → 33198 [SYN, ACK] Seq=0 Ack=1 Win=14480 Len=0 MSS=1440 SACK_PERM TSval=2307889920 TSecr=550051 WS=512
9	0.121265	10.251.23.139	93.17.156.250	TCP	66	33198 → 80 [ACK] Seq=1 Ack=1 Win=5840 Len=0 TSval=550073 TSecr=2307889920
11	0.145144	93.17.156.250	10.251.23.139	TCP	66	80 → 33198 [ACK] Seq=1 Ack=139 Win=15872 Len=0 TSval=2307889926 TSecr=550073
13	0.146311	10.251.23.139	93.17.156.250	TCP	66	33198 → 80 [ACK] Seq=139 Ack=886 Win=8736 Len=0 TSval=550098 TSecr=2307889926
14	0.166927	10.251.23.139	93.17.156.250	TCP	66	33198 → 80 [FIN, ACK] Seq=139 Ack=886 Win=8736 Len=0 TSval=550118 TSecr=2307889926
15	0.188752	93.17.156.250	10.251.23.139	TCP	66	80 → 33198 [FIN, ACK] Seq=886 Ack=140 Win=15872 Len=0 TSval=2307889937 TSecr=550118
16	0.189009	10.251.23.139	93.17.156.250	TCP	66	33198 → 80 [ACK] Seq=140 Ack=887 Win=8736 Len=0 TSval=550141 TSecr=2307889937
35	7.599375	95.136.242.99	216.69.252.100	TCP	82	52573 → 80 [SYN] Seq=0 Win=29200 Len=0 MSS=1452 SACK_PERM TSval=403886 TSecr=0 WS=128
36	7.747665	216.69.252.100	95.136.242.99	TCP	82	80 → 52573 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM TSval=818860362 TSecr=403886 WS=128
37	7.751929	95.136.242.99	216.69.252.100	TCP	74	52573 → 80 [ACK] Seq=1 Ack=1 Win=29312 Len=0 TSval=403932 TSecr=818860362
39	7.902133	216.69.252.100	95.136.242.99	TCP	74	80 → 52573 [ACK] Seq=1 Ack=83 Win=29056 Len=0 TSval=818860516 TSecr=403932
41	7.907512	95.136.242.99	216.69.252.100	TCP	74	52573 → 80 [ACK] Seq=83 Ack=358 Win=30336 Len=0 TSval=403978 TSecr=818860518
42	7.907599	95.136.242.99	216.69.252.100	TCP	74	52573 → 80 [FIN, ACK] Seq=83 Ack=358 Win=30336 Len=0 TSval=403979 TSecr=818860518
43	8.056402	216.69.252.100	95.136.242.99	TCP	74	80 → 52573 [FIN, ACK] Seq=358 Ack=84 Win=29056 Len=0 TSval=818860671 TSecr=403979
44	8.059911	95.136.242.99	216.69.252.100	TCP	74	52573 → 80 [ACK] Seq=84 Ack=359 Win=30336 Len=0 TSval=404024 TSecr=818860671
53	16.237121	2a00:5e80:101:212d:…	2606:f200:0:7:bad:f…	TCP	140	37034 → 80 [SYN] Seq=0 Win=27860 Len=0 MSS=1393 SACK_PERM TSval=406478 TSecr=0 WS=128
54	16.407629	2606:f200:0:7:bad:f…	2a00:5e80:101:212d:…	TCP	142	80 → 37034 [SYN, ACK] Seq=0 Ack=1 Win=28560 Len=0 MSS=1440 SACK_PERM TSval=818869008 TSecr=406478 WS=128
55	16.410925	2a00:5e80:101:212d:…	2606:f200:0:7:bad:f…	TCP	132	37034 → 80 [ACK] Seq=1 Ack=1 Win=27904 Len=0 TSval=406530 TSecr=818869008
57	16.583554	2606:f200:0:7:bad:f…	2a00:5e80:101:212d:…	TCP	134	80 → 37034 [ACK] Seq=1 Ack=83 Win=28672 Len=0 TSval=818869184 TSecr=406530
59	16.587508	2a00:5e80:101:212d:…	2606:f200:0:7:bad:f…	TCP	132	37034 → 80 [ACK] Seq=83 Ack=382 Win=29056 Len=0 TSval=406583 TSecr=818869185
60	16.587683	2a00:5e80:101:212d:…	2606:f200:0:7:bad:f…	TCP	132	37034 → 80 [FIN, ACK] Seq=83 Ack=382 Win=29056 Len=0 TSval=406583 TSecr=818869185
61	16.763408	2606:f200:0:7:bad:f…	2a00:5e80:101:212d:…	TCP	134	80 → 37034 [FIN, ACK] Seq=382 Ack=84 Win=28672 Len=0 TSval=818869359 TSecr=406583
62	16.767158	2a00:5e80:101:212d:…	2606:f200:0:7:bad:f…	TCP	132	37034 → 80 [ACK] Seq=84 Ack=383 Win=29056 Len=0 TSval=406637 TSecr=818869359

7–9 → 10.251.23.139 ↔ 93.17.156.250

35–37 → 95.136.242.99 ↔ 216.69.252.100

53–55 → 2a00:5e80:101:212d:… ↔ 2a00:f200:7:bad:… (IPv6)

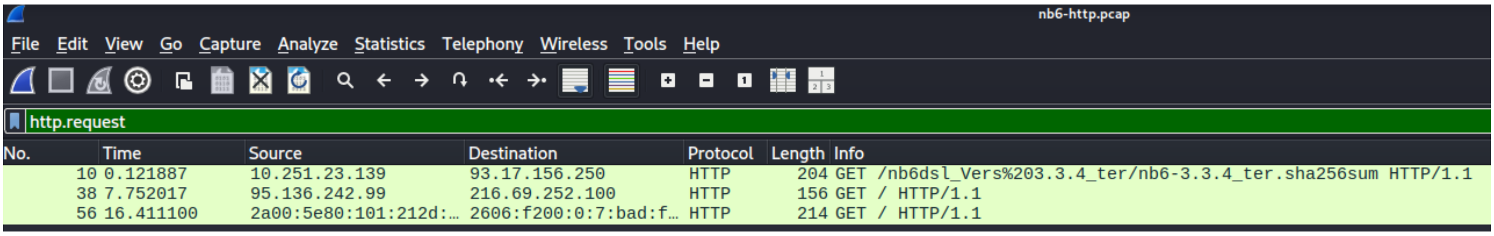
3. Adresowanie w różnych warstwach OSI

Dla żądania HTTP odczytaj:

Źródłowy i docelowy adres MAC, źródłowy i docelowy adres IP, źródłowy i docelowy port TCP.

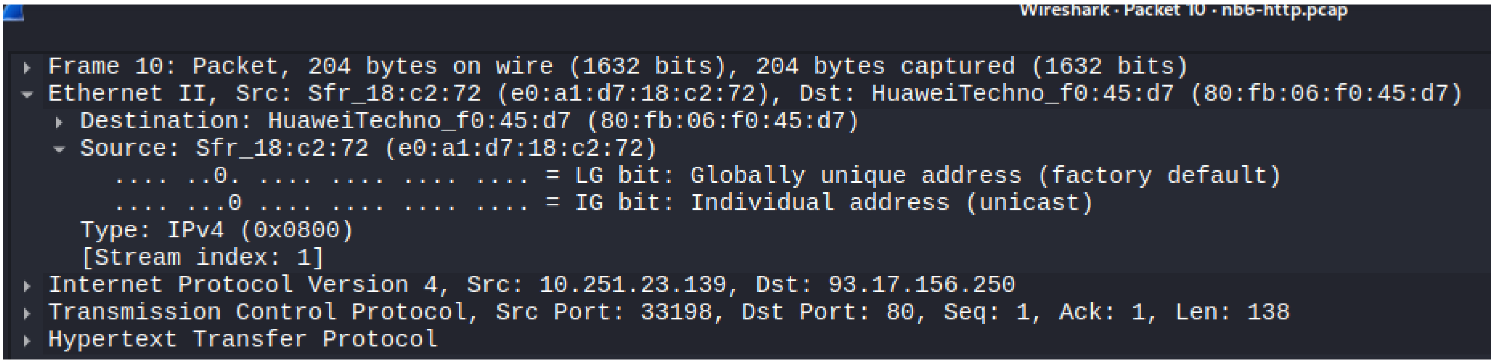
Wyjaśnij, jaką rolę pełni każdy z tych identyfikatorów oraz dlaczego adres MAC wskazuje urządzenie na lokalnym łączu, a adres IP — końcowego odbiorcę pakietu.

Filtr: http.request

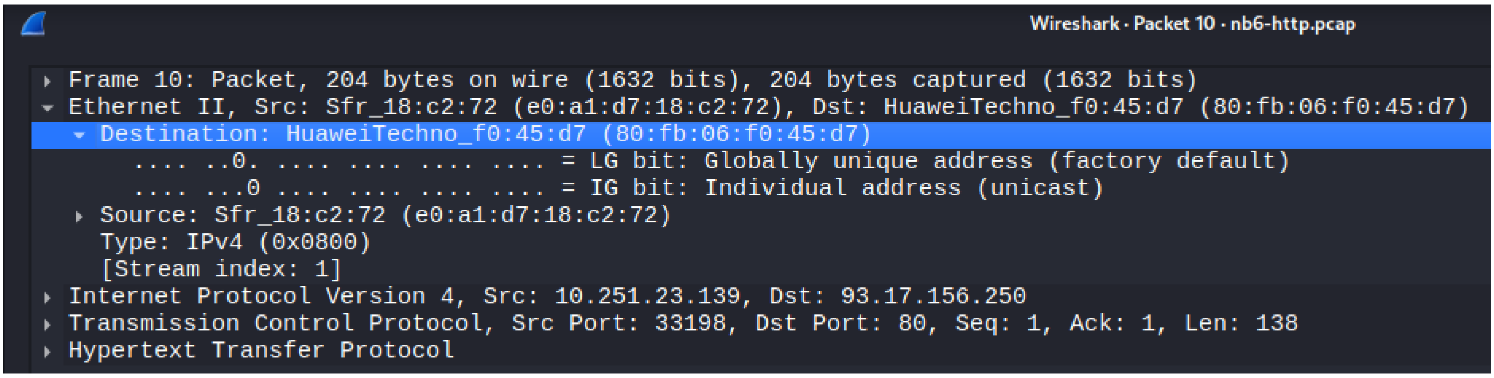


No.	Time	Source	Destination	Protocol	Length	Info
10	0.121887	10.251.23.139	93.17.156.250	HTTP	204	GET /nb6dsl_Vers%203.3.4_ter/nb6-3.3.4_ter.sha256sum HTTP/1.1
38	7.752017	95.136.242.99	216.69.252.100	HTTP	156	GET / HTTP/1.1
56	16.411100	2a00:5e80:101:212d:…	2606:f200:0:7:bad:f…	HTTP	214	GET / HTTP/1.1

Pakiet 10: GET /nb6dsl_Vers%203.3.4_ter/nb6-3.3.4_ter.sha256sum HTTP/1.1



Źródłowy adres MAC: Source: Sfr_18:c2:72 (e0:a1:d7:18:c2:72)



Docelowy adres MAC: Destination: HuaweiTechno_f0:45:d7 (80:fb:06:f0:45:d7)

Źródłowy adres IP: 10.251.23.139
Docelowy adres IP: 93.17.156.250

Źródłowy port TCP: 331198
Docelowy port TCP: 80

Adres MAC
Jest identyfikatorem interfejsu sieciowego na poziomie warstwy 2 (łącza danych). Służy do dostarczenia ramki do następnego urządzenia w lokalnej sieci.

Dlatego docelowy (destination) MAC w pakiecie nie musi być MAC-em serwera 93.17.156.250. Jeśli serwer znajduje się poza lokalną siecią, komputer wysyła ramkę do bramy domyślnej (routera), więc docelowy (destination) adres MAC będzie MAC-em routera.

Adres IP
Działa w warstwie 3 (sieciowej) i identyfikuje źródło oraz końcowy adres, do którego pakiet ma dotrzeć.
W tym przypadku:

10.251.23.139 (źródło) → 93.17.156.250 (cel)

Routery wykorzystują adres IP docelowy, aby zdecydować, dokąd dalej przekazać pakiet.
W przeciwieństwie do adreu MAC, adres IP jest używany do komunikacji między różnymi sieciami.

Port TCP
Działa w warstwie transportowej (warstwa 4). Określa konkretną usługę/proces, do którego ma trafić komunikacja.

33198 → 80

33198 — tymczasowy port klienta

80 — port serwera HTTP

Cały przepływ w skrócie:

MAC → urządzenie na lokalnym odcinku sieci
IP → host docelowy w sieci
Port TCP → konkretna usługa/proces na hoście

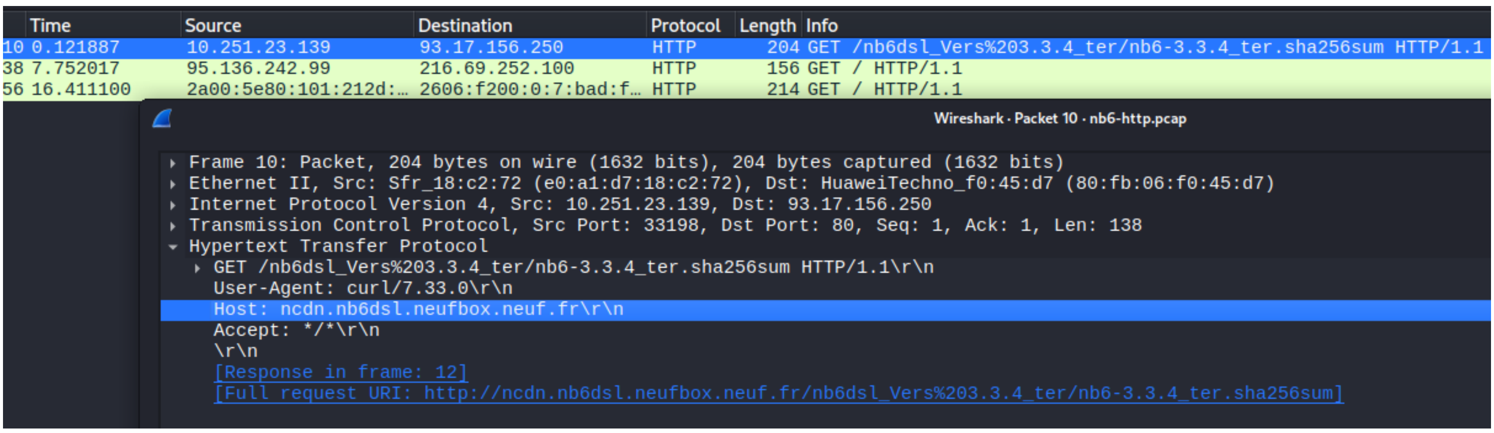
4. **Analiza żądania HTTP**

Otwórz żądanie HTTP i ustal:

jaka metoda została użyta, jaki zasób pobiera klient, jaka jest wersja HTTP, jaka domena znajduje się w nagłówku Host czy treść żądania jest zaszyfrowana.

Filtr: http.request

Dla pakietu No. 10
Metoda: GET
Pobiera zasób: /nb6dsl_Vers%203.3.4_ter/nb6-3.3.4_ter.sha256sum
Wersja HTTP: HTTP/1.1



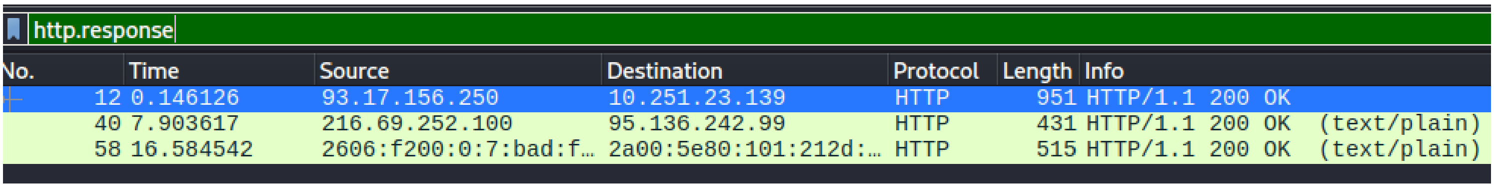
Domena w nagłówku “HOST”: Host: “ncdn.nb6dsl.neufbox.neuf.fr\r\n”
Czy szyfrowana treść: NIE, to HTTP, nie HTTPS, brak TLS

5. Odpowiedź HTTP i zakończenie połączenia

Znajdź odpowiedź serwera i ustal:

jaki kod odpowiedzi zwrócił serwer i co on oznacza, co znajduje się w treści odpowiedzi, które pakiety kończą połączenie TCP i jakie mają flagi.

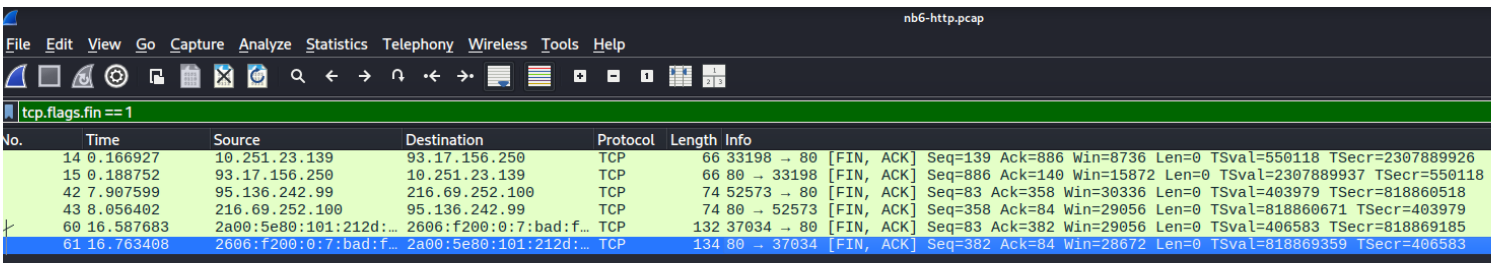
Filtry: http.response oraz tcp.flags.fin == 1



Z pakietu numer 12, widzimy odpowiedz: HTTP/1.1 200 OK
Treść odpowiedzi: “200 OK”
Żądanie zostało pomyślnie obsłużone i serwer zwrócił żądany zasób, o który był poproszony

[Request in frame: 10]
[Request URI: /nb6dsl_Vers%203.3.4_ter/nb6-3.3.4_ter.sha256sum]
[Full request URI: http://ncdn.nb6dsl.neufbox.neuf.fr/nb6dsl_Vers%203.3.4_ter/nb6-3.3.4_ter.sha256sum]
- zapytanie o .sha256sum

Otrzymujemy odpowiedź typu Data



Pakiety 14 i 15 komunikują source i destination między sobą:

```
Wireshark · Packet 14 · nb6-http.pcap

Transmission Control Protocol, Src Port: 33198, Dst Port: 80, Seq: 139, Ack: 886, Len: 0
  Source Port: 33198
  Destination Port: 80
  [Stream index: 0]
  [Stream Packet Number: 8]
▾ [Conversation completeness: Complete, WITH_DATA (31)]
    ..0. .... = RST: Absent
    ...1 .... = FIN: Present
    .... 1... = Data: Present
    .... .1.. = ACK: Present
    .... ..1. = SYN-ACK: Present
    .... ...1 = SYN: Present
    [Completeness Flags: ·FDASS]
  [TCP Segment Len: 0]
  Sequence Number: 139    (relative sequence number)
  Sequence Number (raw): 355167358
  [Next Sequence Number: 140    (relative sequence number)]
  Acknowledgment Number: 886    (relative ack number)
  Acknowledgment number (raw): 3511404818
  1000 .... = Header Length: 32 bytes (8)
▾ Flags: 0x011 (FIN, ACK)
    000. .... .... = Reserved: Not set
    ...0 .... .... = Accurate ECN: Not set
    .... 0... .... = Congestion Window Reduced: Not set
    .... .0.. .... = ECN-Echo: Not set
    .... ..0. .... = Urgent: Not set
    .... ...1 .... = Acknowledgment: Set
    .... .... 0... = Push: Not set
    .... .... .0.. = Reset: Not set
    .... .... ..0. = Syn: Not set
  ▸ .... .... ...1 = Fin: Set
  ▸ [TCP Flags: ·······A···F]
  Window: 4368
  [Calculated window size: 8736]
  [Window size scaling factor: 2]
  Checksum: 0x2190 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
▸ Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
▸ [Timestamps]

000  80 fb 06 f0 45 d7 e0 a1  d7 18 c2 72 08 00 45 b4   ····E···  ···r··E·
```

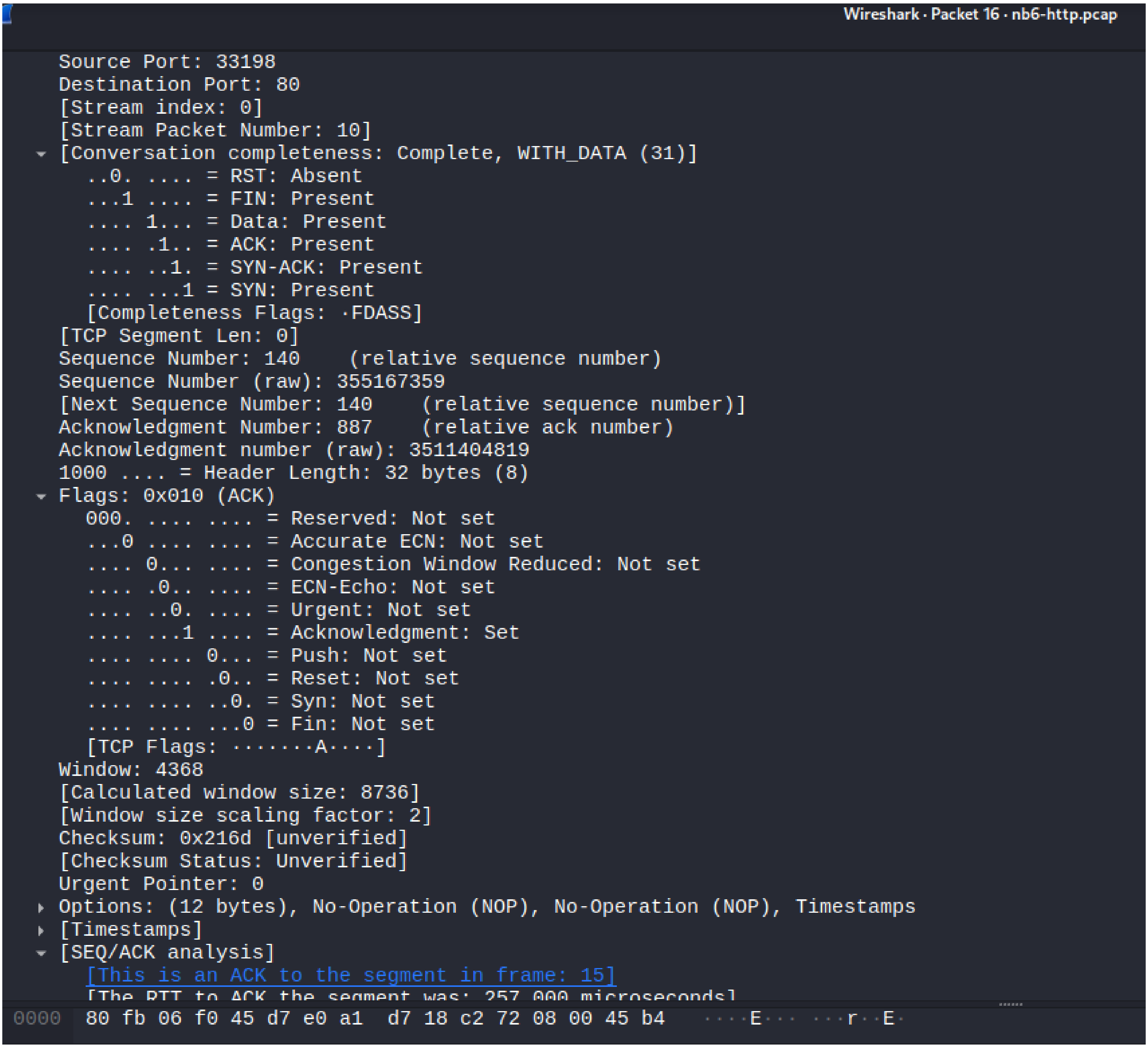
14: src 10.251.23.139 do celu 93.17.156.250 FLAGI **[FIN, ACK]** - klient rozpoczyna zamykanie połączenia

Wireshark · Packet 15 · nb6-http.pcap

```
▾ Transmission Control Protocol, Src Port: 80, Dst Port: 33198, Seq: 886, Ack: 140, Len: 0
    Source Port: 80
    Destination Port: 33198
    [Stream index: 0]
    [Stream Packet Number: 9]
  ▾ [Conversation completeness: Complete, WITH_DATA (31)]
      ..0. .... = RST: Absent
      ...1 .... = FIN: Present
      .... 1... = Data: Present
      .... .1.. = ACK: Present
      .... ..1. = SYN-ACK: Present
      .... ...1 = SYN: Present
      [Completeness Flags: ·FDASS]
    [TCP Segment Len: 0]
    Sequence Number: 886    (relative sequence number)
    Sequence Number (raw): 3511404818
    [Next Sequence Number: 887    (relative sequence number)]
    Acknowledgment Number: 140    (relative ack number)
    Acknowledgment number (raw): 355167359
    1000 .... = Header Length: 32 bytes (8)
  ▾ Flags: 0x011 (FIN, ACK)
      000. .... .... = Reserved: Not set
      ...0 .... .... = Accurate ECN: Not set
      .... 0... .... = Congestion Window Reduced: Not set
      .... .0.. .... = ECN-Echo: Not set
      .... ..0. .... = Urgent: Not set
      .... ...1 .... = Acknowledgment: Set
      .... .... 0... = Push: Not set
      .... .... .0.. = Reset: Not set
      .... .... ..0. = Syn: Not set
    ▸ .... .... ...1 = Fin: Set
    ▸ [TCP Flags: ·······A···F]
    Window: 31
    [Calculated window size: 15872]
    [Window size scaling factor: 512]
    Checksum: 0x3275 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ▸ Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  ▸ [Timestamps]
```

15: src 93.17.156.250 do celu 10.251.23.139 FLAGI **[FIN, ACK]** - serwer również zamyka swoje połączenie

```
14 0.166927   10.251.23.139   93.17.156.250   TCP   66 33198 → 80 [FIN, ACK] Seq=139 Ack=886 Win=8736 Len=0 TSval=550118 TSecr=2307889926
15 0.188752   93.17.156.250   10.251.23.139   TCP   66 80 → 33198 [FIN, ACK] Seq=886 Ack=140 Win=15872 Len=0 TSval=2307889937 TSecr=550118
16 0.189009   10.251.23.139   93.17.156.250   TCP   66 33198 → 80 [ACK] Seq=140 Ack=887 Win=8736 Len=0 TSval=550141 TSecr=2307889937
```



```
Wireshark · Packet 16 · nb6-http.pcap
    Source Port: 33198
    Destination Port: 80
    [Stream index: 0]
    [Stream Packet Number: 10]
  ▾ [Conversation completeness: Complete, WITH_DATA (31)]
      ..0. .... = RST: Absent
      ...1 .... = FIN: Present
      .... 1... = Data: Present
      .... .1.. = ACK: Present
      .... ..1. = SYN-ACK: Present
      .... ...1 = SYN: Present
      [Completeness Flags: ·FDASS]
    [TCP Segment Len: 0]
    Sequence Number: 140    (relative sequence number)
    Sequence Number (raw): 355167359
    [Next Sequence Number: 140    (relative sequence number)]
    Acknowledgment Number: 887    (relative ack number)
    Acknowledgment number (raw): 3511404819
    1000 .... = Header Length: 32 bytes (8)
  ▾ Flags: 0x010 (ACK)
      000. .... .... = Reserved: Not set
      ...0 .... .... = Accurate ECN: Not set
      .... 0... .... = Congestion Window Reduced: Not set
      .... .0.. .... = ECN-Echo: Not set
      .... ..0. .... = Urgent: Not set
      .... ...1 .... = Acknowledgment: Set
      .... .... 0... = Push: Not set
      .... .... .0.. = Reset: Not set
      .... .... ..0. = Syn: Not set
      .... .... ...0 = Fin: Not set
      [TCP Flags: ·······A····]
    Window: 4368
    [Calculated window size: 8736]
    [Window size scaling factor: 2]
    Checksum: 0x216d [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ▸ Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  ▸ [Timestamps]
  ▾ [SEQ/ACK analysis]
      [This is an ACK to the segment in frame: 15]
      [The RTT to ACK the segment was: 257.000 microseconds]
0000  80 fb 06 f0 45 d7 e0 a1  d7 18 c2 72 08 00 45 b4   ····E···  ···r··E·
```

Na końcu pakiet numer 16: src 10.251.23.139 do celu 93.17.156.250 **FLAGA [ACK]** - czyli potwierdza FLAGĘ [FIN] serwera. " [This is an ACK to the segment in frame: 15]"